

Lecture-2

Mathematics 1 (15B11MA111)

CO [C105.1]

Preliminaries

Module: Partial differentiation

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics”
fifth edition, Narosa publishing house, 2016.

Topics to be covered

➤ **Function of two variables**

➤ **Neighborhood of a point**

➤ **Limit**

❖ **Definition**

❖ **Examples**

❖ **Assignment**

➤ Function of two variables

Let $z = f(x, y)$ be a function of two variables then the set of points (x, y) for which z or $f(x, y)$ be defined is called domain of z and the corresponding values of z is called range of z .

For example: (i) Let $z = \sqrt{1 - x^2 - y^2}$ and z is real.

Then $1 - x^2 - y^2 \geq 0$, i.e. $x^2 + y^2 \leq 1$, which is its domain and the range is the set of all real positive numbers.

(ii) Let $z = \frac{1}{x-y}$.

Here domain is the set of all points for which $x \neq y$ and the range is $\mathbb{R}$.

Distance between two points

The distance between the points $A(x_0, y_0)$ and $B(x_1, y_1)$ is given

$$\text{as } d(A, B) = \sqrt{(x_1 - x_0)^2 + (y_1 - y_0)^2}$$

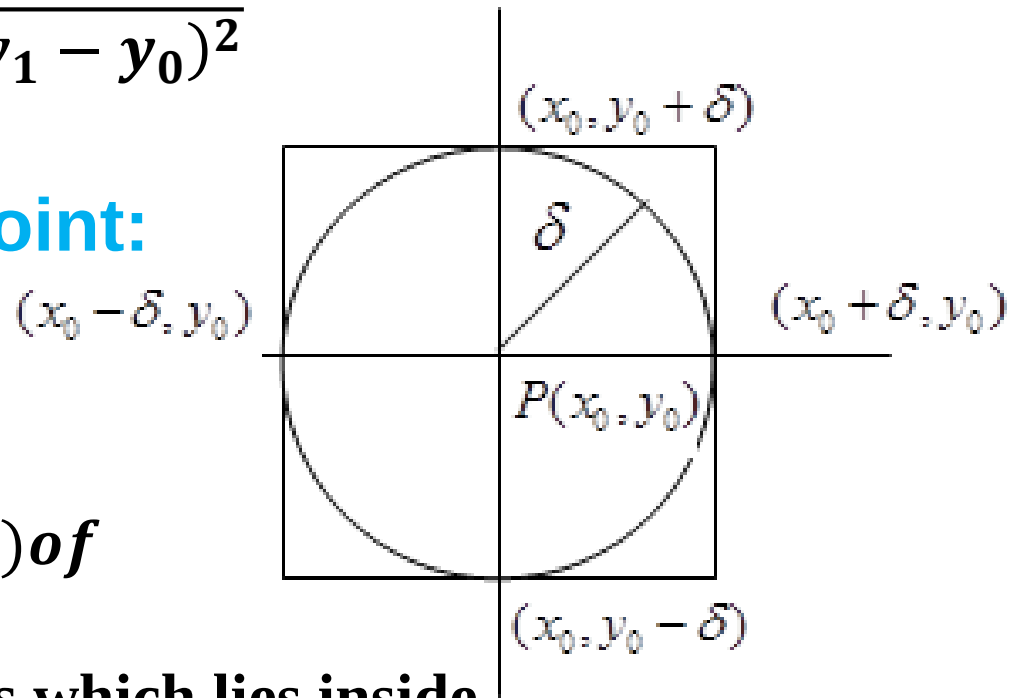
➤ Neighborhood of a point:

Let $P(x_0, y_0)$ be a point in 2D

Space. The δ – neighborhood of

$P(x_0, y_0)$ is the set of all points which lies inside

the circle of radius δ with center at $P(x_0, y_0)$.



Hence, $N_\delta(P) = \{(x, y): \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta\}$

Since $|x - x_0| \leq \sqrt{(x - x_0)^2 + (y - y_0)^2}$ and

$$|y - y_0| \leq \sqrt{(x - x_0)^2 + (y - y_0)^2},$$

The neighborhood can be written as

$$N_\delta(P) = \{(x, y): |x - x_0| < \delta \text{ and } |y - y_0| < \delta\}$$

= set of all points lies inside a square of size 2δ

➤ Limit

❖ Definition

Let $z = f(x, y)$ be a function of two variables in 2D space. Let $P(x_0, y_0)$ be a point in it. If for a given real number $\epsilon > 0$, however small, we can find a real number $\delta > 0$ such that for every point (x, y) in the δ -neighborhood of $P(x_0, y_0)$ $|f(x, y) - L| < \epsilon$, whenever $\sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$ then $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = L$, provided that the limit exists.

If $f(x, y)$ is not defined at the point $P(x_0, y_0)$ then

$$|f(x, y) - L| < \epsilon, \text{ whenever } 0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$$

❖ Remarks

- i.* $\lim_{(x,y) \rightarrow (x_0, y_0)} f(x, y)$, if it exists then it is unique.
- ii.* Since in 2D plane there are infinite number of ways (path) to reach from $(x, y) \rightarrow (x_0, y_0)$ therefore the limit should be same (unique) along all the possible paths. Limit should be independent of path.
- iii.* (r, θ) approach: let $x = r \cos \theta, y = r \sin \theta$ then $x^2 + y^2 = r^2$ and $\theta = \tan^{-1} \frac{y}{x}$ and now the limit is defined as
$$\lim_{r \rightarrow 0} |f(r \cos \theta, r \sin \theta) - L| < \epsilon, \text{ whenever } r < \delta,$$
 independent of θ .

Properties

If $\lim_{(x,y) \rightarrow (x_0, y_0)} f(x, y) = L_1$, and $\lim_{(x,y) \rightarrow (x_0, y_0)} g(x, y) = L_2$,

i. $\lim_{(x,y) \rightarrow (x_0, y_0)} [kf(x, y)] = kL_1$, for any real k .

ii. $\lim_{(x,y) \rightarrow (x_0, y_0)} [f(x, y) \pm g(x, y)] = L_1 \pm L_2$.

iii. $\lim_{(x,y) \rightarrow (x_0, y_0)} [f(x, y)g(x, y)] = L_1L_2$.

iv. $\lim_{(x,y) \rightarrow (x_0, y_0)} [f(x, y)/g(x, y)] = \frac{L_1}{L_2}$, $L_2 \neq 0$.

Examples: Using $\varepsilon - \delta$ approach, show that

(i) $\lim_{(x,y) \rightarrow (2,1)} [3x + 4y] = 10$. (ii) $\lim_{(x,y) \rightarrow (1,1)} [x^2 + 2y] = 3$.

Solution

(i) Here $f(x, y) = 3x + 4y$ and it is defined at (2,1).

$$\begin{aligned}|f(x, y) - 10| &= |3x + 4y - 10| = |3(x - 2) + 4(y - 1)| \\ &\leq 3|x - 2| + 4|y - 1|.\end{aligned}$$

Taking $|x - 2| < \delta$ and $|y - 1| < \delta$, we get $|f(x, y) - 10| < 7\delta < \epsilon$, which is true when $\delta < \epsilon/7$, Hence

$$\lim_{(x,y) \rightarrow (2,1)} [3x + 4y] = 10.$$

(ii) Here $\lim_{(x,y) \rightarrow (1,1)} [x^2 + 2y] = 3$ is defined at (1,1).

$$\begin{aligned}|f(x, y) - 3| &= |x^2 + 2y - 3| \\ &= |(x - 1 + 1)^2 + 2(y - 1 + 1) - 3|\end{aligned}$$

$$\begin{aligned}
 &= |(x-1)^2 + 2(x-1) + 2(y-1)| \\
 &\leq |x-1|^2 + 2|x-1| + 2|y-1|.
 \end{aligned}$$

Taking $|x-1| < \delta$ and $|y-1| < \delta$, we get $|f(x,y) - 3| < \delta^2 + 4\delta < \varepsilon$, or $\delta < \sqrt{\varepsilon + 4} - 2$.

Hence $\lim_{(x,y) \rightarrow (1,1)} [x^2 + 2y] = 3$

Example:

(iii) Using $\varepsilon - \delta$ approach, show that $\lim_{(x,y) \rightarrow (0,0)} \left(\frac{xy}{\sqrt{x^2 + y^2}} \right) = 0$

Solution: Here $f(x,y)$ is not defined at $(0,0)$

$$\left| \frac{xy}{\sqrt{x^2 + y^2}} - 0 \right| = \left| \frac{xy}{\sqrt{x^2 + y^2}} \right| \leq \frac{1}{2} \frac{(x^2 + y^2)}{\sqrt{x^2 + y^2}}$$

$$= \frac{1}{2} \sqrt{x^2 + y^2} < \epsilon, \text{ since } |xy| \leq (x^2 + y^2)/2.$$

Choosing $\delta < 2\epsilon$, we get

$$\left| \frac{xy}{\sqrt{x^2 + y^2}} - 0 \right| < \epsilon, \text{ whenever } 0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta.$$

Hence

$$\lim_{(x,y) \rightarrow (0,0)} \left(\frac{xy}{\sqrt{x^2 + y^2}} \right) = 0$$

Example:

Show that the following limits do not exist

$$(i) \lim_{(x,y) \rightarrow (0,0)} \left(\frac{xy}{x^2 + y^2} \right) \quad (ii) \lim_{(x,y) \rightarrow (0,0)} \left(\frac{x + \sqrt{y}}{x^2 + y} \right)$$

(iii) $\lim_{(x,y) \rightarrow (0,0)} \left(\frac{x^3 y}{x^6 + y^2} \right)$

Solutions: (i) Consider a path $y = mx$,

then $\lim_{(x,y) \rightarrow (0,0)} \left(\frac{xy}{x^2 + y^2} \right) = \lim_{x \rightarrow 0} \frac{mx^2}{(1+m^2)x^2} = \frac{m}{(1+m^2)}$ which depends on m .

Limit does not exist.

(ii) Consider a path $y = mx^2$, we get $\lim_{(x,y) \rightarrow (0,0)} \left(\frac{x + \sqrt{y}}{x^2 + y} \right) = \lim_{x \rightarrow 0} \frac{1 + \sqrt{m}}{(1+m)x} = \infty$, not finite, limit does not exist.

(iii) Consider a path $y = mx^3$, we see that the limit depends on m .
Hence limit does not exist.

Assignment

(i) Using $\delta - \varepsilon$ approach, establish the limit

$$\lim_{(x,y) \rightarrow (2,1)} (x^2 + 2x - y^2) = 7.$$

(ii) Determine the following limits if they exist

$$\lim_{(x,y) \rightarrow (1,-1)} \frac{x^3 - y^3}{x - y}.$$

Thank you